SDST3600/STAT3600 Linear Statistical Analysis
Comprehensive Practice Paper

This paper is reorganized to follow the style of the class test / midterm.
It includes:
1. Original assignment questions that were required for submission;
2. Consolidated lecture-slide examples, regrouped by dataset;
3. Tutorial exercises, excluding the small number of R-specific subparts.

General notes:
1. Present numerical answers in 4 decimal places unless otherwise stated.
2. Show key intermediate steps for hand-calculation questions.
3. Use R only where the original assignment/tutorial question clearly requires data fitting.
4. No solutions are included in this practice paper.

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SECTION A ASSIGNMENT SUBMIT QUESTIONS

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Question 1 Assignment 1, Q15 (submit parts a-e)
Consider a linear regression model when Y is regressed on X for 6 observations. It is given that:

$$\sum x_i = 9, \quad \sum y_i = 24.4$$
$$\sum x_i^2 = 31, \quad \sum y_i^2 = 116.72$$
$$\sum x_i y_i = 52.4$$

(a) Calculate the least squares estimates of the intercept and the slope.
(b) Calculate the sum of squared errors and the unbiased estimate of the variance of the error.
(c) Calculate the standard errors of the estimated intercept and slope.
(d) Construct a 90% confidence interval for the intercept and the slope.
(e) Predict an individual value of Y when X = 1. Construct a 99% prediction interval for the prediction.

Question 2 Assignment 1, Q16 (submit parts a-g)
Four observations of two variables are given as follows:

X :	-2	-1	0	1
Y :	1.9	2.0	0.9	0.1

Consider a linear regression model when Y is regressed on X. It is given that for any 2 x 2 matrix [[a, b], [c, d]],

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

(a) Write down the data matrix.
(b) Calculate the least squares estimates of the intercept and the slope.

(c) Calculate the sum of squared errors and the unbiased estimate of the variance of the error.

(d) Estimate the covariance matrix of the estimators of the intercept and the slope.

(e) Calculate the standard errors of the estimated intercept and slope.

(f) Test at the 5% level of significance whether the slope is -1.

(g) Estimate the mean of Y when X = -1.5. Construct a 99% confidence interval for the estimate.

Question 3 Assignment 1, Q17 (submit parts a-e)

The present study aimed to assess the health risk associated with magnetic fields generated by high-voltage power lines. The magnetic flux density (density, micro T) was measured around high-voltage 63 kV power lines at various distances (distance, m) from the transmission lines. The data are stored in 'leukemia.csv'.

(a) Plot magnetic flux density against distance. Is a linear regression appropriate?

(b) Fit a simple linear regression model to the dataset and plot the fitted regression line on the graph obtained in (a). Report the least squares estimates of the regression coefficients.

(c) Interpret the intercept and the slope quantitatively.

(d) Test at the 5% level of significance whether distance is effective in explaining the variation in magnetic flux density.

(e) Predict the magnetic flux density of an individual location with 30 m from the power lines. Construct a 95% prediction interval.

Question 4 Assignment 2, Q5 (submit full question)

Show that the multiple coefficient of determination in a multiple regression model is the square of the coefficient of correlation between Y and Y-hat.

Question 5 Assignment 2, Q7 (submit parts a-i)

You are given the following matrices computed for a regression analysis

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

$$X'X = \begin{bmatrix} 20 & 7 & 7 \\ 7 & 15 & 0 \\ 7 & 0 & 15 \end{bmatrix}$$

$$X'Y = [14, 22, 19]'$$

$$(X'X)^{-1} = \frac{1}{3030} \begin{bmatrix} 225 & 105 & 105 \\ 105 & 251 & 49 \\ 105 & 49 & 251 \end{bmatrix}$$

$$Y'Y = 70$$

The matrices are properly ordered according to the regression function above.

(a) Calculate the LSE of the regression coefficients. Describe the effects of the regressors on the response variable quantitatively.
(b) Calculate SSE and MSE.
(c) Estimate the covariance matrix of the estimates.
(d) Calculate the standard error of the estimates.
(e) Calculate a 95% confidence interval for β_1 and β_2, respectively.
(f) Construct an ANOVA table. Test whether there is a regression of Y on X_1 and X_2 at the 5% level of significance.
(g) Calculate R^2. Comment on the fitness of the model.
(h) Test at the 5% level of significance whether $\beta_1 + \beta_2 = 0$ and $\beta_1 - \beta_2 = 1$, simultaneously.
(i) Estimate the means of Y for two cases where $(X_1, X_2) = (1, 1)$ and $(-1, -1)$. Construct (i) a 90% simultaneous interval based on Bonferroni's method and (ii) a 95% simultaneous interval based on Scheffe's method.

Question 6 Assignment 3, Q7 (submit part d only)
Consider a regression analysis of Y on X_1 - X_4 for 60 observations. The SSE for various sub-models are given below.

None	173.8385	X1, X2	94.3016	X1, X2, X3	93.0149
X1	172.0857	X1, X3	168.8508	X1, X2, X4	50.1196
X2	94.3081	X1, X4	148.8978	X1, X3, X4	147.2592
X3	170.8197	X2, X3	93.0152	X2, X3, X4	50.0693
X4	151.6323	X2, X4	50.1507	X1, X2, X3, X4	50.0328
X3, X4	150.1505				

Determine the subset of variables that is selected by the forward selection method, based on the entry level F = 4. Show your steps. Report SSR, MSE and F-value at each step. Report the selected regressors.

Question 7 Assignment 3, Q10 (submit parts a-e and g)
Six observations of a dependent variable X and a factor A are given as follows.

```
A : 1  1  2  2  3  3
X : 3  2  1  3  7  6
```

Consider a one-way classification model.

(a) Write down a cell-means model for the analysis.
(b) Estimate the parameters in (a).
(c) Obtain the fitted values for X.

(d) Compile the ANOVA table.
(e) Conduct a size 0.05 test to determine whether or not the means for X for the three levels of A differ.
(g) By the Bonferroni's method, construct the simultaneous 95% confidence intervals for the pairwise comparisons among the three levels. Comment on the comparison.

A Student's t table is given as:

df	0.05	0.025	0.0167	0.0083
2	4.30	6.21	7.65	10.89
3	3.18	4.18	4.86	6.23
4	2.78	3.50	3.96	4.85
5	2.57	3.16	3.53	4.22

Question 8 Assignment 3, Q11 (submit parts a-f and i)
The following data of X are given for three levels of factor A and 2 levels of factor B.
There are two observations for each treatment.

	A: 1	2	3
B=1	2, 3	1, 1	7, 4
B=2	0, 0	3, 1	8, 4

(a) Calculate and plot the means for X for the six treatments. Does it appear interaction effects between factors A and B? Explain.
(b) Write down a factor-effects model for the 2-way classification model with interaction.
(c) Estimate the main effects of factors A and B and the effects of interaction, respectively.
(d) Compile the ANOVA table.
(e) Test the interaction effects at the 5% level of significance. State null and alternative hypothesis, decision rule and conclusion.
(f) Test the main effects for factor A at the 5% significance level. Write down the null and alternative hypothesis, decision rule and conclusion.
(i) Consider the 3 pairwise comparisons of the 3 levels of factor A within each level of factor B. Construct Bonferroni's simultaneous 95% confidence intervals for all six pairwise comparisons simultaneously. Comment on the comparisons.

A Student's t table is given as:

df	0.05	0.025	0.0083	0.0042
4	2.78	3.50	4.85	5.89
5	2.57	3.16	4.22	4.98
6	2.45	2.97	3.86	4.49
7	2.36	2.84	3.64	4.17
8	2.31	2.75	3.48	3.96

Question 9 Assignment 4, Q4 (submit full question)
Six observations of a response variable Y and a treatment A are given as follows.

A : 1 1 1 2 3 3

Y : 2 1 2 7 4 3

(a) Use level 3 of A as the reference level and the zero constraint (ii) in 7.2.1, state the regression model equivalent to an ANOVA model. State the data matrix X of the regression model.
(b) Calculate X'X, X'Y and Y'Y.
(c) It is given that
[[6,3,1],[3,3,0],[1,0,1]]^(-1) = (1/6) * [[3,-3,-3],[-3,5,3],[-3,3,9]].
Calculate the LSE of the regression coefficients.
(d) Compile the ANOVA table. Test the effects of A at the 5% level of significance. State the hypotheses, decision rule and conclusion.
(e) Conduct a t test for each of the coefficients at the 5% level of significance. Interpret the results.
(f) Express mean Y for each level of A in terms of the parameters in (a).
(g) Express the pairwise comparisons of mean Y for the 3 levels of A in terms of the parameters in (a). Thus, make all pairwise comparisons between the treatment effects; use a 90% confidence level by the Bonferroni's method. Comment on the comparisons.

Question 10 Assignment 4, Q7 (submit full question)
Five observations of a response variable Y, a treatment A and a covariate X are given as follows.

A :	1	2	2	3	3
X :	3	2	1	4	5
Y :	31	22	11	43	53

It is given that
[[5,1,2,15],[1,1,0,3],[2,0,2,3],[15,3,3,55]]^(-1)
= (1/4) * [[83,-29,-56,-18],[-29,15,20,6],[-56,20,40,12],[-18,6,12,4]].

(a) Use level 3 of A as the reference level. Using the zero constraint (ii) in 7.2.1, state the regression model equivalent to an ANCOVA model. State the data matrix X.
(b) Calculate the LSE for the regression coefficients and SSE in (a).
(c) State the reduced model for testing the effects of A and calculate SSE for the reduced model.
(d) Test the effects of A; use alpha = 0.05. State the hypotheses, decision rule and conclusion.
(e) Use the Bonferroni's method at 90% confidence level, conduct a pairwise comparison of mean Y among the three levels of A.
(f) Construct a 95% confidence interval for the average of two mean Y when the values of A and X are:

(A, X) = (1, 1), (3, 4)

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SECTION B CONSOLIDATED LECTURE-SLIDE EXAMPLES

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Question 11 Lecture Example: Westwood Company Data (SLR)
The data are:

Production run lot size X : 30 20 60 80 40 50 60 30 70 60
Man-hours Y   : 73 50 128 170 87 108 135 69 148 132

Consider the simple linear regression model of man-hours on lot size
.

(a) Calculate x-bar, y-bar, Sxx and Sxy.
(b) Obtain the least squares estimates of the intercept and slope.
(c) Calculate SSE and MSE.
(d) Estimate the covariance matrix of the estimators.
(e) Construct 95% confidence intervals for beta_0 and beta_1.
(f) Test H0: beta_1 = 0 against H1: beta_1 != 0 at the 5% level.
(g) Estimate the mean man-hours when lot size is 55 and construct a 95% confidence interval.

Question 12 Lecture Example: Cholesterol Data (SLR + simultaneous inference)
The data are given in 24 pairs:

X : 46 20 52 30 57 25 28 36 22 43 57 33 22 63 40 48 28 49 52 58 29 34 24 50
Y : 3.5 1.9 4.0 2.6 4.5 3.0 2.9 3.8 2.1 3.8 4.1 3.0 2.5 4.6 3.2 4.2 2.3 4.0 4.3 3.9 3.3 3.2 2.5 3.3

Let cholesterol level Y be regressed on age X.

(a) Obtain the fitted regression line.
(b) Calculate SSE and MSE.
(c) Construct 95% confidence intervals for beta_0 and beta_1.
(d) Test H0: beta_1 = 0 at the 5% level.
(e) Construct simultaneous 95% confidence intervals for the mean cholesterol levels at ages 50 and 60 using:
i. Bonferroni's method;
ii. Scheffe's method.
(f) Briefly compare the widths of the intervals obtained in (e).

Question 13 Lecture Example: Environmental Data (MLR)
Consider the multiple regression of ozone Y on radiation X1, temperature X2 and wind X3 for 30 days. The following matrices are given:

```
X'X =
[[30, 5733, 2085, 354.7],
 [5733, 1463179, 411076, 67085.7],
 [2085, 411076, 147243, 24523],
```

```
[354.7, 67085.7, 24523, 4583.39]]

X'Y = [83.84, 17097.38, 5953.47, 973.64]'

(X'X)^(-1) =
[[ 2.8559226,  0.0006173, -0.0353507, -0.0409097],
 [ 0.0006173,  0.0000033, -0.0000181,  0.0000000],
 [-0.0353507, -0.0000181,  0.0005339,  0.0001439],
 [-0.0409097,  0.0000000,  0.0001439,  0.0026139]]
```

(a) Calculate the least squares estimates of the regression coefficients.

(b) Write down the fitted model and interpret each slope coefficient quantitatively.

(c) State the null hypothesis for testing whether there is a regression relation and explain how the ANOVA table should be constructed.

(d) Explain how to obtain the covariance matrix of beta-hat.

(e) Estimate the expected ozone level for a day with radiation 250, temperature 80 and wind speed 10. State the form of the confidence interval needed for the mean response.

Question 14 Lecture Example: Car Mileage Data (one-way ANOVA)
The following data give the number of miles/gallon recorded when four versions of a new car were test driven on a few separate occasions:

```
A: 2000 cc automatic          30 26 27 29
B: 2000 cc manual estate      28 29 28 31
C: 2000 cc manual saloon      29 32 27 31 30
D: 1300 cc manual saloon      45 35 45 37 40
```

(a) Estimate the treatment mean response for each car version.

(b) Compile the one-way ANOVA table.

(c) Test at the 5% level whether car version has a significant effect on mileage.

(d) Construct Bonferroni simultaneous confidence intervals for all six pairwise comparisons among the treatment means.

(e) State clearly which version appears most petrol-friendly and justify your answer statistically.

Question 15 Lecture Example: Education and Income Data (randomized block design)
Monthly income data (in dollars) were collected from four companies, with two technicians observed at each of three education levels:

```
Education level 1 (pre-high school):
    Company 1: 8000, 7500
    Company 2: 6500, 9000
    Company 3: 10100, 11050
    Company 4: 7200, 7000

Education level 2 (high school):
    Company 1: 10000, 9100
    Company 2: 8800, 9000
    Company 3: 15000, 14300
    Company 4: 6500, 7500
```

Education level 3 (college):
Company 1: 17000, 16500
Company 2: 10000, 11000
Company 3: 24000, 25500
Company 4: 9300, 8900

Treat education level as the factor of interest and company as a block.

(a) Write down the randomized block design model.
(b) Explain why blocking by company is reasonable here.
(c) Compile the ANOVA table for testing education effects.
(d) Test whether education level has a significant effect on income at the 1% level.
(e) Let L = (theta_1 + theta_2)/2 - theta_3. Estimate L and construct a 95% confidence interval for L.

SECTION C TUTORIAL EXERCISES (R-SPECIFIC SUBPARTS REMOVED)

Question 16 Tutorial 1: Matrix and Distribution Theory

Given

```
A = [[3, 4, -1, 5],
     [7,-2,  2, 1]]

B = [[ 3, -1],
     [ 5,  2],
     [-3, -2],
     [ 4,  0]]
```

(a) What are the dimensions of A and B respectively?
(b) Does A + B exist?
(c) Calculate AB and (AB)'.
(d) Calculate B'A'.
(e) Are there any relationship between the results of (c) and (d)? What does it imply?
(f) Do A^(-1) and B^(-1) exist?
(g) Calculate (AB')^(-1) if it exists.

Question 17 Tutorial 2: Multivariate normal distributions

Let

[Y1, Y2]' ~ N_2([1, 1]', [[1, 1], [1, 4]]).

(a) What is the distribution of:
i. Y1 - 2Y2
ii. Z = [Z1, Z2]' = [Y1 + Y2, 2Y1 - Y2]' ?
(b) Write down the covariance between Z1 and Z2.

Also, let X1, ..., Xn be n iid samples from the same distribution N_d(0, Omega_X) and v be a known vector in R^d.

(c) What is the distribution of X1'v?
(d) Let A = (X1, ..., Xn) in R^(d x n). What is the distribution of A'v?
(e) Let S = (1/n) A A'. Show that S is an unbiased estimator of Cov(X).

Question 18 Tutorial 3: SLR prediction and confidence region
A study investigates the relationship between the average number of hours people spent on using the internet per day and their age. The data are:

Hours	3.3	2.5	4.7	0.25	3.0	1.03	1.75	8.24
Age	34	47	25	56	29	62	41	24

Use Age as the explanatory variable and Hours as the response variable.

(a) Find n, X-bar, Y-bar, Sxx, Sxy and Syy.
(b) Compute the least squares estimates of beta_0 and beta_1.
(c) Find an unbiased estimate for sigma^2.

Question 19 Tutorial 6: Midterm revision
(a) Suppose X ~ N_n(0, I). The sample mean X-bar and sample variance S^2 are defined as usual. Show that X-bar and S^2 are independent using the independence theorem.
(b) Find the distribution of S^2 using Cochran's theorem.

Let X and Y be independent standard normal random variables. Define

Q1 = 0.36 X^2 + 0.64 Y^2 + 0.96 XY
Q2 = 0.64 X^2 + 0.36 Y^2 - 0.96 XY

(c) Show that Q1 and Q2 can be written in quadratic forms and specify the symmetric matrices A1 and A2.
(d) Find the distributions of Q1 and Q2.

For the following simple linear regression data:

Ratio	0.05	0.5	1.25	2.1	2.5
Desorbed	0.1	0.8	1.81	3.20	4.01

Consider a simple linear regression of Desorbed on Ratio.

(e) Write down the regression model and state clearly the assumptions.
(f) Estimate and interpret the regression coefficients.
(g) Find the predicted values and residuals.
(h) Test whether Ratio has any effect on Desorbed based on the t test at the 5% level.
(i) Predict the mean Desorbed level when Ratio is 2.3.

Question 20 Tutorial 8, 9 and 10: Classification, two-way ANOVA and GLM
Part I. One-way ANOVA (Tutorial 8) A clinical psychologist compared three methods for reducing hostility levels in university students. The post-treatment

HLT scores are:

Method 1: 80, 92, 87, 83
Method 2: 70, 81, 78, 82
Method 3: 63, 76, 70

(a) State the null and alternative hypotheses.
(b) Calculate Syy, SSE and SStr.
(c) Produce the ANOVA table.
(d) Perform the ANOVA test at alpha = 0.05.
(e) Perform a t test to compare the means of groups 1 and 2 at alpha = 0.05.

Part II. Two-way ANOVA (Tutorial 9)
The average costs of U.S. electric power sources are:

	Traditional		Alternative	
Year	Coal	Gas	Solar	Wind
1985	9.0	11.5	19.5	11.5
1993	5.5	4.5	9.0	6.0
2000	4.5	3.5	5.5	4.5

Use time (3 levels) and type of source (2 levels) as factors.

(f) Set up a two-way ANOVA table to test at the 5% significance level for interaction effects. Explain your findings by means of a visual display.
(g) After testing for interaction effects, test for the two main effects.
(h) Comment on whether alternative-energy costs appear to be approaching those of gas and coal.

Part III. General linear model (Tutorial 10)
Spending data from three regions are given:

North : 3346 3114 4669 4888 3547 3782 4247 3982 3568 3155 2967 3285

Central: 4517 4349 3731 2853 2533 2305 3124 3429
Others : 3402 2297 2932 4123

(i) Write down the general linear model and state clearly the assumptions.
(j) Recode the factor levels to dummy variables.
(k) Estimate the treatment mean response by means of the parameter estimates of the general linear model.
(l) Test the treatment effect by using the F test.
(m) The researcher assumes that northern citizens spend on average more than the average of those in other regions. Do you agree at the 95% confidence level?

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END OF PRACTICE PAPER

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